

Understanding Frequency Ancillary Services

Grid Codes, Frequency-Control Hierarchy,
Resource Activation, and Grid-Feasible Flexibility

Study notes based on Section 3.2 of Miah et al. (2026)

Key idea

- **Frequency indicates the balance between active-power generation and demand:**

$$P_{\text{generation}} \approx P_{\text{demand}} \Rightarrow f \approx f_0$$

- If **generation is smaller than demand**, frequency falls:

$$P_{\text{generation}} < P_{\text{demand}} \Rightarrow f \downarrow$$

- If **generation is larger than demand**, frequency rises:

$$P_{\text{generation}} > P_{\text{demand}} \Rightarrow f \uparrow$$

- After a disturbance, the power system activates **different forms of frequency support at different times**.
- **Fast resources respond first**; slower resources then take over to **restore frequency and replace the reserves** that were activated.

1 Why Does Frequency Change?

The electrical power system needs to continuously maintain approximately

$$P_{\text{generation}} = P_{\text{demand}}.$$

For a 50 Hz power system, normal operation can be represented as

$$f \approx 50 \text{ Hz}.$$

Suppose that a large generator suddenly disconnects.

Then,

$$P_{\text{generation}} < P_{\text{demand}}.$$

There is suddenly not enough generation to supply the demand.

As a result,

$$\text{frequency starts to fall.}$$

For example,

$$50.00 \rightarrow 49.90 \rightarrow 49.80 \rightarrow 49.70 \text{ Hz}.$$

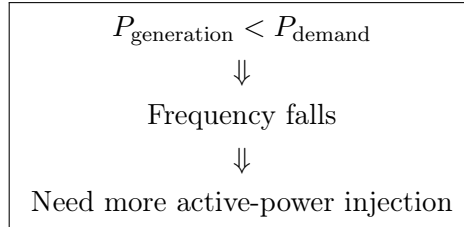
The opposite situation can also occur.

If

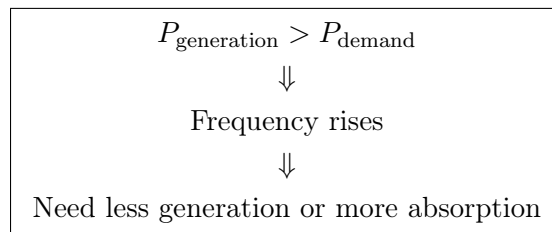
$$P_{\text{generation}} > P_{\text{demand}},$$

there is excessive generation and frequency rises.

Therefore,



whereas



2 Simple BESS Interpretation

A battery energy storage system (BESS) can react in both directions.

For a low-frequency event,

$$f < f_0,$$

the battery can discharge and provide additional active power.

For a high-frequency event,

$$f > f_0,$$

the battery can charge and absorb active power.

In plain language,

Frequency too low $\rightarrow$ BESS discharges

and

Frequency too high $\rightarrow$ BESS charges.
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Here,

$$f_0$$

is the nominal frequency, such as 50 Hz or 60 Hz.

3 What Happens After a Frequency Disturbance?

One of the most important concepts is that frequency support does not come from only one action. Different actions occur at different times.

A simplified sequence is

Disturbance → Inertia → FFR / Primary → Secondary → Replacement

Each stage has a different job.

3.1 Illustrative frequency trajectory

The following graph is an illustrative example. The numerical values are chosen only to explain the concept.

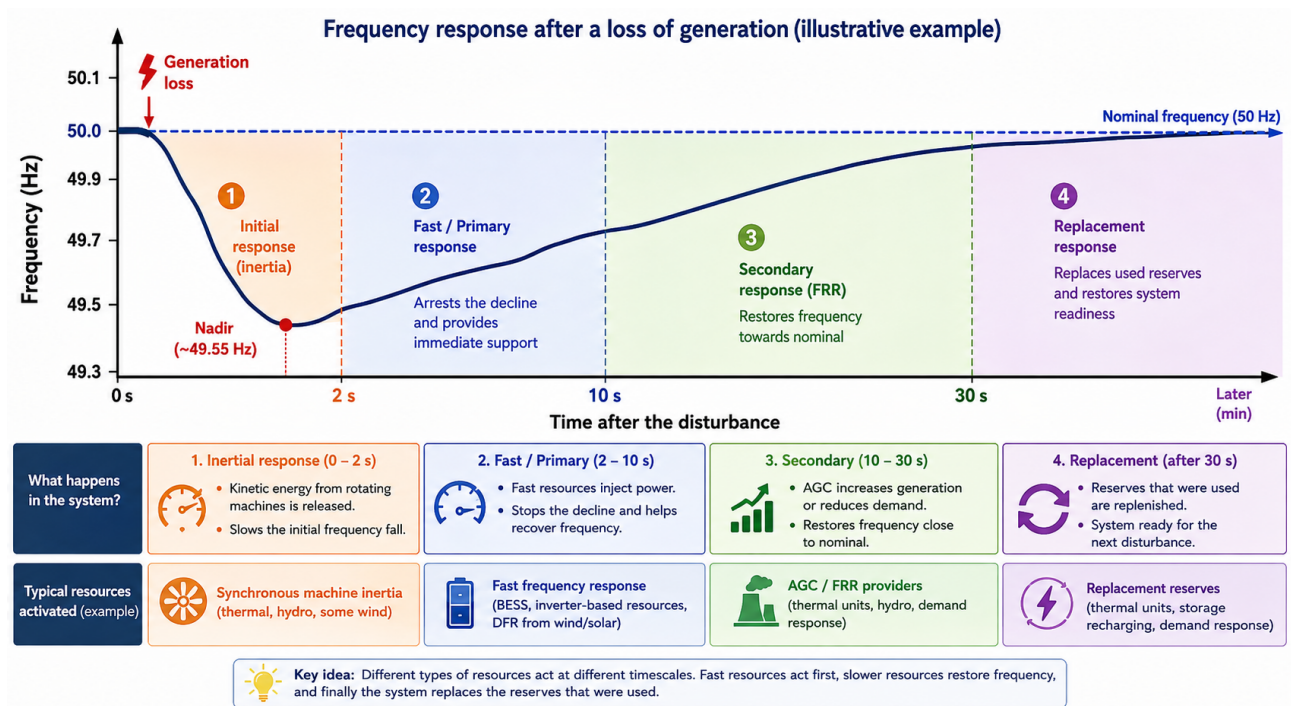


Figure 1: Illustrative evolution of frequency after a loss of generation. The values are explanatory rather than grid-code values.

4 The Frequency-Control Hierarchy

This is the concept that is sometimes explained as “different generators taking over at different stages.” That interpretation is useful, but it requires one important clarification.

Important distinction

There is **not necessarily one fixed technology assigned to each frequency-control stage**. Instead, every service has technical requirements such as:

- how quickly the resource must react;
- how much power it can provide;
- how long the response must be sustained;
- whether the response is automatic or centrally commanded;
- whether sufficient energy is available;
- whether the resource is qualified to provide that service.

Different technologies can therefore provide the same service if they satisfy its requirements.

4.1 Visual explanation

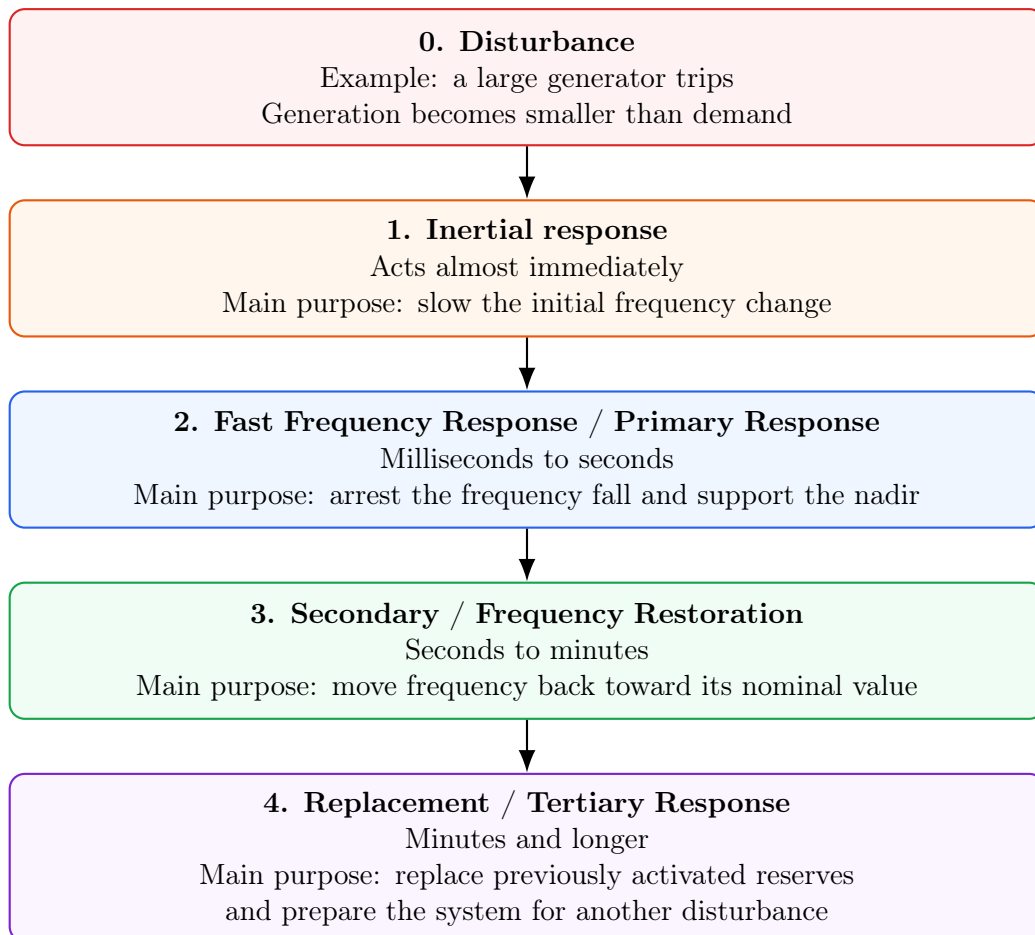


Figure 2: Simplified hierarchy of frequency-control actions after a disturbance.

5 Which Resources Can Act at Each Stage?

Different technologies are suitable for different parts of the response because their physical characteristics are different.

The following table should therefore be interpreted as **typical examples**, not as a rigid rule.

Stage	Typical speed	Typical resources	Main job
Inertia	Immediate	Synchronous generators; rotating machines; synthetic/virtual inertia from properly controlled inverter-based resources	Slow the initial rate of frequency change
Fast Frequency Response	Milliseconds to a few seconds	BESS, fast demand response, converter-based resources, some wind and other inverter-based resources	Very quickly inject or absorb active power
Primary response / FCR	Seconds	Hydro units, gas turbines, governor-controlled thermal units, BESS, responsive loads, qualified inverter-based resources	Arrest the frequency deviation and stabilize frequency
Secondary response / FRR	Seconds to minutes	AGC-controlled hydro, gas generation, BESS, flexible generation and qualified aggregated resources	Restore frequency and scheduled system balance
Replacement / tertiary reserve	Minutes and longer	Dispatchable hydro, gas and thermal generation, demand response, storage and other scheduled reserves	Replace previously activated reserves and prepare for the next event

Table 1: Illustrative mapping between frequency-control stages and resource types. Several technologies can participate in more than one stage.

Key idea

- The **fast resource** does not necessarily provide the service for the entire event.
- A **BESS** can inject power almost immediately after the frequency disturbance.
- After a short time, **slower dispatchable resources**, such as hydro or conventional generators, can increase their generation.
- These slower resources then **take over a larger share of the power imbalance**.
- This allows the BESS to **reduce or stop its discharge**, helping preserve or recover its **state of charge (SOC)**.

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6 A Simple Numerical Example of the Resource Sequence

Consider a simplified **50 Hz power system**. A generator producing

$$100 \text{ MW}$$

suddenly trips. Therefore, the system experiences an initial active-power deficit of

$$\boxed{100 \text{ MW}}.$$

The following simplified sequence illustrates how different frequency-control resources can progressively respond to the disturbance.

Step 1: Generator trip and initial frequency decline

At

$$t = 0,$$

the 100 MW generator disconnects.

- Generation becomes approximately **100 MW smaller than demand**.
- System frequency immediately starts to fall.
- Rotating synchronous machines provide **inertial response**.
- This response is automatic and does not require a new dispatch instruction.
- Its main role is to **slow the initial rate of frequency decline**.

At this stage, the active-power deficit is still approximately

$$\boxed{100 \text{ MW}}.$$

Step 2: Fast resources respond

After a fraction of a second, suppose fast BESS resources inject

$$30 \text{ MW}.$$

The remaining imbalance becomes

$$100 - 30 = 70 \text{ MW}.$$

Therefore,

$$\boxed{\text{Remaining imbalance} = 70 \text{ MW}}.$$

- The BESS reacts very quickly.
- Its additional active-power injection helps **arrest the frequency decline**.
- The BESS does not necessarily need to sustain this contribution for the entire event.

Step 3: Primary frequency-response resources increase output

A few seconds later, suppose primary frequency-response resources provide

$$40 \text{ MW.}$$

The total controlled response is now

$$30 + 40 = 70 \text{ MW.}$$

Therefore, the remaining imbalance becomes

$$100 - 70 = 30 \text{ MW.}$$

Thus,

$\text{Remaining imbalance} = 30 \text{ MW.}$

- Primary-response resources take over part of the imbalance.
- Frequency is no longer falling as severely.
- The fast BESS response can progressively be reduced as other resources contribute.

Step 4: Secondary control restores the remaining balance

Secondary-control resources then provide the remaining

$$30 \text{ MW.}$$

The total response becomes

$$30 + 40 + 30 = 100 \text{ MW.}$$

Therefore, the original 100 MW generation loss has now been compensated:

$\text{Remaining imbalance} = 0 \text{ MW.}$
--

- Secondary resources help restore the generation–demand balance.
- Frequency moves back toward its nominal value.
- Some of the earlier fast response can now be released.

Step 5: Replacement resources restore system readiness

After the immediate frequency event has been controlled, slower resources may be dispatched to replace the reserves that were activated.

- Fast reserves can return to their normal operating condition.
- A BESS can reduce its discharge and preserve or recover its SOC.
- The system rebuilds reserve capacity so that it is prepared for another disturbance.

Key idea

The simplified numerical sequence can be summarized as:

Generator loss	: -100 MW
BESS response	: +30 MW
Primary response	: +40 MW
Secondary response	: +30 MW
Total response	: +100 MW

Therefore,

$$-100 + 30 + 40 + 30 = 0 \text{ MW.}$$

In this simplified example, different resources progressively contribute to replace the lost generation and restore the active-power balance.

Important distinction

- The numerical values are used only to explain the sequence.
- Actual frequency response is a **continuous dynamic process**.
- The different frequency-control services can **overlap in time**.
- One service does not need to completely finish before the next one begins.
- The important idea is that **fast resources act first**, while slower and more sustainable resources progressively take over the response.

7 RoCoF

RoCoF means

Rate of Change of Frequency.

It describes how quickly frequency is changing:

$$\text{RoCoF} = \frac{df}{dt}.$$

The unit is normally

Hz/s.

Simple numerical example

Suppose frequency changes from

50.0 Hz

to

49.5 Hz

in

0.5 s.

Then,

$$\text{RoCoF} = \frac{49.5 - 50.0}{0.5} = -1 \text{ Hz/s.}$$

The negative sign means that frequency is falling.

8 Frequency Nadir

The **frequency nadir** is the lowest frequency reached following a frequency-drop event.

Suppose frequency follows

50.00 → 49.80 → 49.60 → 49.55 → 49.65 → 49.80 Hz.

The nadir is

$$f_{\text{nadir}} = 49.55 \text{ Hz.}$$

The frequency has stopped falling at this point and starts recovering.

Key idea

Fast frequency response is especially important before the nadir.

Its purpose is to inject active power quickly enough to help arrest the frequency decline before frequency reaches a dangerous value.

9 FFR — Fast Frequency Response

FFR means

Fast Frequency Response.

It is a very fast active-power response following a frequency event. If a grid code specifies

$$t_{\text{FFR}} < 1 \text{ s,}$$

this does **not** mean that the battery performs an optimization every second. It means:

After a qualifying frequency disturbance occurs, the resource must provide its required response within the specified response time.

Do not confuse these concepts

Grid-code response time and **optimization time resolution** are different concepts.

A BESS scheduling model may use intervals of several minutes while the physical BESS controller can respond to a frequency event in less than one second.

10 Primary Frequency Response and Frequency Containment Reserve

10.1 What do PFR and FCR mean?

- **PFR** means

Primary Frequency Response.

- **FCR** means

Frequency Containment Reserve.

The terminology is not identical in every country, but both concepts are related to the **first controlled response after a frequency deviation**. Their main purpose is:

to stop the frequency from continuing to move away from its nominal value

In other words, this response helps **contain or arrest the frequency deviation**.

10.2 When does this response act?

Suppose a large generator trips.

Then,

$$P_{\text{generation}} < P_{\text{demand}},$$

and the frequency starts to fall.

The sequence is approximately:

Generator trip → frequency falls → PFR/FCR activates → additional active power is provided

The objective is not necessarily to bring frequency immediately back to

50 Hz.

Instead, the first objective is to

stop the frequency decline.

This means helping the system reach a stable point after the disturbance.

10.3 How does the resource know how much power to provide?

A common approach is to use the **measured local frequency**.

The resource continuously observes

$$f.$$

If

$$f = f_0,$$

there is no frequency deviation and no additional primary response is needed.

If

$$f < f_0,$$

frequency is too low, so the resource should increase active-power injection.

If

$$f > f_0,$$

frequency is too high, so the resource should reduce injection or absorb active power.

This can be summarized as:

$f < f_0$	$\Rightarrow$	inject more active power
$f = f_0$	$\Rightarrow$	no additional response
$f > f_0$	$\Rightarrow$	reduce injection or absorb power

10.4 Simplified droop relationship

A simple way to represent this behavior is

$$\Delta P = K_f(f_0 - f)$$

where:

- f_0 = **nominal frequency**, for example

$$f_0 = 50 \text{ Hz};$$

- f = **measured system frequency**;
- $f_0 - f$ = **frequency deviation**;
- K_f = **frequency-response gain**, which determines how strongly the resource changes its active power;
- ΔP = **change in active power** provided by the resource.

10.5 How to interpret the equation

Consider the equation

$$\Delta P = K_f(f_0 - f).$$

If frequency is below nominal,

$$f < f_0,$$

then

$$f_0 - f > 0.$$

Therefore,

$$\Delta P > 0.$$

This means that the resource provides **additional active power**.

For a BESS, this normally means:

the battery discharges more.

On the other hand, if

$$f > f_0,$$

then

$$f_0 - f < 0,$$

and therefore

$$\Delta P < 0.$$

For a BESS, this can mean:

the battery charges or reduces its discharge.

10.6 Simple numerical example

Assume a 50 Hz system:

$$f_0 = 50 \text{ Hz.}$$

After a generation loss, the measured frequency becomes

$$f = 49.80 \text{ Hz.}$$

The frequency deviation is

$$f_0 - f = 50 - 49.80 = 0.20 \text{ Hz.}$$

Assume an illustrative frequency-response gain of

$$K_f = 5 \text{ MW/Hz.}$$

Then,

$$\Delta P = K_f(f_0 - f)$$

becomes

$$\Delta P = 5(0.20) = 1 \text{ MW.}$$

Therefore, the resource increases its active-power injection by

$$\Delta P = 1 \text{ MW.}$$

For a BESS, this means that the battery discharges an additional 1 MW to help contain the frequency decline.

10.7 The main idea

Key idea

- PFR/FCR acts after a frequency deviation has started.
- The response is usually based on the **measured frequency**.
- If frequency falls, participating resources provide more active power.
- If frequency rises, participating resources reduce active-power injection or absorb power.
- The main objective is to **arrest or contain the deviation**, not necessarily to restore frequency completely to its nominal value.
- The later restoration of frequency is performed by slower frequency-restoration mechanisms.

11 Secondary Frequency Control

Primary response may arrest the frequency fall without restoring the system exactly to

$$50 \text{ Hz.}$$

A second level is therefore needed. This is commonly called **secondary frequency control** or **frequency restoration**.

FRR means

$$\text{Frequency Restoration Reserve.}$$

Its purpose is to move the system frequency back toward its nominal value and restore the scheduled system balance.

11.1 AGC

AGC means

Automatic Generation Control.

Unlike a local autonomous fast-frequency response, AGC is a centralized control action. The system operator detects an imbalance and changes the setpoints of participating resources.

In simplified form,

System measurement $\rightarrow$ AGC $\rightarrow$ new resource setpoints.

12 Replacement Reserve (RR)

RR means

Replacement Reserve.

Replacement Reserve is a **slower reserve service** that is activated after the faster frequency-control resources have already responded.

Its main purpose is to

take over part of the sustained power response and release faster reserves.

12.1 Why is Replacement Reserve needed?

After a frequency disturbance:

1. **Fast resources respond first** to help contain the frequency deviation.
2. **Frequency-restoration resources** then help restore the generation–demand balance.
3. If these resources remain activated for a long time, their available reserve may be reduced. For a BESS, for example, continuous discharge reduces its **state of charge (SOC)**.
4. **Replacement Reserve is activated later** to take over part of this sustained power requirement.
5. The faster resources can then be **released and made available for another disturbance**.

12.2 What can provide Replacement Reserve?

An important distinction is:

Replacement Reserve is a service, not a specific technology.

A physical resource can provide RR if it satisfies the required **activation time, power capability, duration, and applicable grid-code or market requirements**.

Possible providers can include, depending on the system and market:

- **hydroelectric generators;**
- **gas-fired or other dispatchable generators;**
- **energy storage**, including BESSs.

Therefore, it is better to say

A hydro generator provides Replacement Reserve

rather than

Replacement Reserve is a hydro generator.

12.3 Simple numerical example

Suppose a BESS is initially providing

30 MW

following a frequency disturbance. Later, a hydro generator is activated to provide Replacement Reserve. The handover could be illustrated as:

	BESS	Hydro providing RR
Initially	30 MW	0 MW
Handover 1	20 MW	10 MW
Handover 2	10 MW	20 MW
After replacement	0 MW	30 MW

During the handover,

$$P_{\text{BESS}} + P_{\text{hydro}} = 30 \text{ MW.}$$

The system therefore continues receiving the required 30 MW, but the BESS is progressively released. This helps the BESS **preserve or recover its SOC** and restores to respond to a future event.

12.4 Main idea

Key idea

- RR is a **service**, not a particular generator technology.
- It acts **later than the faster frequency-control reserves**.
- Resources such as **hydro, gas generation, demand response, or storage** may provide RR if they satisfy the service requirements.
- Its role is to **take over sustained power delivery and release previously activated reserves**.
- In the German example reported in the studied paper, FCR, FRR, and RR are required to activate within approximately **30 s, 5 min, and 15 min**, respectively.

13 Grid-Code Comparison from the Studied Paper

The reviewed paper compares RoCoF, frequency-nadir management, and FFR requirements in several regions.

Region	RoCoF withstand	Frequency-nadir focus	FFR
Australia	Up to 1 Hz/s for 500 ms on mainland	Prevent nadir from reaching critical values, especially under low-inertia operation	< 1 s
China	Up to 1.5 Hz/s	Maintain nadir above approximately 49.2 Hz	< 2 s
ENTSO-E	Up to 2 Hz/s for 500 ms	DERs participate in frequency-nadir management	< 1 s
France	Up to 1.5 Hz/s	Maintain nadir approximately within 49.5–50 Hz	< 2 s
Germany	Up to 2.5 Hz/s for 200 ms	Prevent nadir from falling below approximately 49.5 Hz	< 1 s
India	Up to 1 Hz/s	DERs help maintain nadir above approximately 49.5 Hz	< 2 s
United Kingdom	Up to 1 Hz/s for 500 ms	DER response arrests the frequency drop before nadir	< 1 s
United States	Approximately 0.5–1 Hz/s	Maintain frequency-nadir stability across different regions	1–2 s

Table 2: Plain-language summary of the frequency requirements reported in Table 6 of Miah et al. (2026).

14 The UK Dynamic Containment Example

The paper gives a useful example from Great Britain. For Dynamic Containment, it reports approximately

$$\boxed{\text{maximum initiation time} = 0.5 \text{ s}}$$

and

$$\boxed{\text{maximum full-delivery time} = 1 \text{ s.}}$$

The reported delivery duration is

$$15 \text{ min.}$$

Notice that these are three different characteristics:

$$\boxed{\text{Start quickly} + \text{Reach the required power quickly} + \text{Sustain it long enough.}}$$

15 Response Speed and Response Duration Are Different

A resource can be **very fast** but unable to sustain its response for a long time. Another resource can be **slower** but capable of providing power for a much longer period. Therefore, two different characteristics must be distinguished:

$$\boxed{\text{Response speed} \neq \text{Response duration}}$$

- **Response speed:** how quickly the resource can change its active power after a frequency event.
- **Response duration:** how long the resource can maintain that power response.

Simple comparison

Consider two illustrative resources:

	Resource A: BESS	Resource B: Slower generator
Power response	1 MW	1 MW
Time to reach response	0.5 s	30 s
Sustainable duration	5 min	Several hours
Main strength	Very fast response	Long-duration support
Typical role	Early frequency support	Later sustained response

Table 3: Illustrative comparison between response speed and response duration.

How to interpret the example

1. **Immediately after the disturbance**, the BESS is more useful because it can reach

1 MW

in only

0.5 s.

2. However, suppose the BESS can maintain this response for only

5 min.

Its stored energy therefore limits how long it can continue providing the service.

3. The slower generator needs

30 s

to reach the same 1 MW response, so it is less suitable for the very first instant of the event.

4. Once activated, however, the generator can maintain the response for several hours.
5. The slower resource can therefore progressively take over part of the sustained power requirement, allowing the BESS to reduce its discharge.

Main idea

Key idea

- A **fast response** is important immediately after the disturbance.
- A **long-duration response** is important if the power imbalance persists.
- A BESS can be excellent for the **early response**, but its available energy and SOC may limit how long it can sustain that response.
- A slower dispatchable resource may be less useful in the first second, but more suitable for **sustained support**.
- This is one reason why frequency control uses several response layers rather than relying on a single resource.

Fast resource → acts first → slower sustainable resource takes over

16 Headroom for Frequency Response

To provide frequency-response services, a resource must have **available flexibility beyond its current operating point**. This available margin is called

headroom.

For an aggregated BESS, it is useful to distinguish three different requirements:

Power headroom + Energy headroom + Network headroom

All three must be sufficient for the requested service.

1. Power Headroom

Can the BESS increase or decrease its active power right now?

Suppose the maximum BESS discharge power is

$$P^{\text{dis},\text{max}} = 2 \text{ MW}.$$

If the battery is already discharging at

$$P^{\text{dis}} = 2 \text{ MW},$$

then its additional upward capability is

$$R_{\text{power}}^{\uparrow} = P^{\text{dis},\text{max}} - P^{\text{dis}} = 2 - 2 = 0 \text{ MW}.$$

Therefore,

$$R_{\text{power}}^{\uparrow} = 0$$

and the battery cannot increase its active-power injection further.

If instead the battery is discharging at only

$$P^{\text{dis}} = 0.5 \text{ MW},$$

then

$$R_{\text{power}}^{\uparrow} = 2 - 0.5 = 1.5 \text{ MW}.$$

Thus,

$$R_{\text{power}}^{\uparrow} = 1.5 \text{ MW}.$$

The battery now has 1.5 MW of additional upward power capability.

2. Energy Headroom

Having enough power capability is not sufficient. The BESS must also have enough stored energy to **sustain the requested response**. Therefore, energy headroom answers:

$$\text{Can the BESS maintain this additional power for the required duration?}$$

For an upward frequency response, the battery needs sufficient stored energy above its minimum SOC.

$$SOC > SOC^{\min}.$$

For a downward response, in which the BESS may need to charge, it also needs sufficient empty storage capacity:

$$SOC < SOC^{\max}.$$

Simple numerical example

Suppose a BESS can provide

$$1.5 \text{ MW}$$

of additional discharge power. However, because of its current SOC, it can sustain this response for only

$$2 \text{ min.}$$

If the ancillary service requires the response to be sustained for

$$15 \text{ min},$$

then the full 1.5 MW cannot be offered for that service. Therefore,

$$\text{Power capability exists, but energy headroom is insufficient.}$$

3. Network Headroom

Even if the BESS has sufficient power and energy capability, the distribution network must also be able to accommodate the resulting change in power flow. Network headroom answers:

Can this additional BESS response be delivered at the PCC without violating network constraints?

The requested response must respect, for example:

- voltage limits;
- branch-current limits;
- transformer loading limits;
- inverter capability limits;
- other distribution-network operating constraints.

This is the component that connects directly with the **Feasible Operating Region (FOR)**. If the requested PCC operating point is outside

$$\mathcal{F}_k,$$

then the response is not network-feasible, even if the batteries themselves could physically provide it.

16.1 Putting the Three Headrooms Together

The actual frequency-response capability is therefore limited by the most restrictive condition. Conceptually,

$$R_k^\uparrow = \min \left\{ R_{\text{power},k}^\uparrow, R_{\text{energy},k}^\uparrow, R_{\text{network},k}^\uparrow \right\}.$$

This means that the largest technically valid response is not determined only by the battery rating. It depends on:

BESS power capability + available energy + distribution-grid feasibility.

16.2 Simple Combined Example

Suppose the current operating condition gives:

$$R_{\text{power}}^\uparrow = 1.5 \text{ MW},$$

$$R_{\text{energy}}^\uparrow = 1.2 \text{ MW},$$

and

$$R_{\text{network}}^\uparrow = 1.0 \text{ MW}.$$

Then the actual upward response that can be safely offered is

$$R^\uparrow = \min \{1.5, 1.2, 1.0\}.$$

Therefore,

$$R^\uparrow = 1.0 \text{ MW.}$$

Although the batteries could physically increase their output by 1.5 MW, the distribution network allows only 1.0 MW under the current operating condition.

Connection with the PhD framework

For the PhD framework, this distinction is important. The BESS determines the available **power and energy flexibility**, while the FOR determines whether that flexibility can be safely delivered through the distribution grid to the PCC. Therefore,

$$\text{DER flexibility} \neq \text{grid-feasible flexibility at the PCC.}$$

The flexibility that can actually be offered to the upstream system is the portion that satisfies both the BESS constraints and the distribution-network constraints.

““latex

17 Connection with the Feasible Operating Region

The **Feasible Operating Region (FOR)** describes the active- and reactive-power exchanges that the distribution system can achieve at the PCC while satisfying the modeled network and DER constraints.

At time k ,

$$(P_{\text{PCC}}, Q_{\text{PCC}}) \in \mathcal{F}_k$$

where $\mathcal{F}_k$ is the FOR calculated for the current network conditions and BESS state.

For frequency-response provision, the important question is:

$$\text{How far can the PCC operating point move in } P \text{ while remaining inside the FOR?}$$

17.1 Simple Numerical Example

Assume the current PCC operating point is

$$P_0 = -0.5 \text{ MW},$$

where negative active power represents **export to the upstream grid**.

For the current reactive-power condition Q_0 , suppose the FOR allows active power down to

$$P_{\min} = -3.0 \text{ MW}.$$

Therefore, the PCC can move from

$$-0.5 \text{ MW}$$

to

$$-3.0 \text{ MW}$$

without leaving the feasible region.

The corresponding network-feasible upward support is

$$R_{\text{network}}^{\uparrow} = P_0 - P_{\min}.$$

Substituting the values,

$$R_{\text{network}}^{\uparrow} = -0.5 - (-3.0) = 2.5 \text{ MW}.$$

Thus,

$$R_{\text{network}}^{\uparrow} = 2.5 \text{ MW}.$$

In plain language, the distribution system may increase its export to the upstream grid by up to **2.5 MW** from the present operating point, from the network-feasibility perspective.

17.2 Visual Interpretation

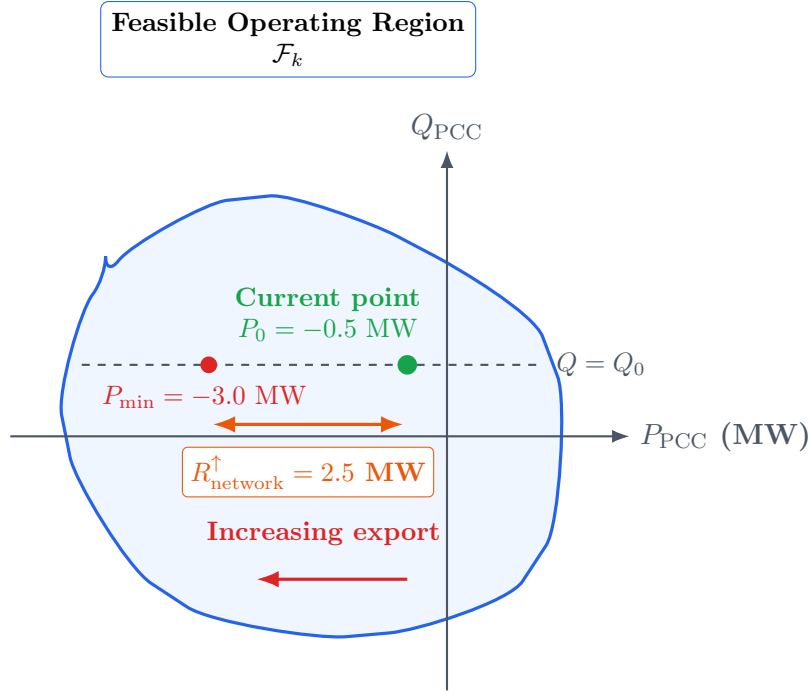


Figure 3: Illustrative connection between the current PCC operating point and the network-feasible active-power headroom. For the same reactive-power condition Q_0 , the PCC can move from $P_0 = -0.5$ MW to $P_{\min} = -3.0$ MW while remaining inside the FOR.

17.3 How to Interpret the Figure

1. The **green point** represents the current PCC operating condition:

$$(P_0, Q_0).$$

2. The blue area represents the **FOR**, containing the PCC operating points that satisfy the modeled network and resource constraints.
3. For the same reactive-power level Q_0 , the PCC can move toward greater export until reaching the FOR boundary at

$$P_{\min} = -3.0 \text{ MW}.$$

4. The distance between the current point and this boundary is the available **network headroom** in that direction:

$$R_{\text{network}}^{\uparrow} = 2.5 \text{ MW}.$$

17.4 Connection with Frequency Response

Suppose a low-frequency event occurs and additional active-power support is required from the distribution system. The current PCC exchange is

$$P_0 = -0.5 \text{ MW}.$$

If the frequency-response controller requires an additional

$$\Delta P = -1.5 \text{ MW},$$

the new PCC exchange becomes

$$P_{\text{PCC}}^{\text{service}} = -0.5 - 1.5 = -2.0 \text{ MW}.$$

Since

$$-2.0 \text{ MW}$$

is still between the current point and the feasible boundary

$$P_{\min} = -3.0 \text{ MW},$$

the requested response remains inside the illustrative FOR. Therefore,

$$(P_{\text{PCC}}^{\text{service}}, Q_0) \in \mathcal{F}_k$$

from the network-feasibility perspective.

Connection with the PhD framework

- The **current operating point** tells us where the distribution system is operating now.
- The **FOR boundary** tells us how far the PCC exchange can move before a modeled network or resource constraint becomes binding.
- The distance between them provides a measure of **network-feasible headroom**.
- For frequency response, this headroom indicates how much additional active-power support may be delivered to the upstream system without leaving the FOR.
- The FOR alone does **not** guarantee that the response can be reached fast enough or sustained for the required duration. Power capability, SOC, and dynamic response must also be considered.

18 But the FOR Alone Is Not Enough

This is an important point for frequency ancillary services.

Suppose the FOR says that

$$2.5 \text{ MW}$$

of additional export is network-feasible. This does not automatically mean that the distribution system can offer

$$2.5 \text{ MW}$$

of a sub-second frequency service. The resources must also be able to reach that response quickly enough. Therefore, for a particular ancillary service s , we can think conceptually about

$$R_{s,k}^{\uparrow} = \min \left\{ R_{\text{network},k}^{\uparrow}, R_{\text{power},k}^{\uparrow}, R_{\text{dynamic},s,k}^{\uparrow}, R_{\text{energy},s,k}^{\uparrow} \right\}.$$

where:

- $R_{\text{network},k}^{\uparrow}$: maximum response allowed by the distribution network;
- $R_{\text{power},k}^{\uparrow}$: maximum response allowed by BESS/inverter power ratings;
- $R_{\text{dynamic},s,k}^{\uparrow}$: amount that can physically be reached within the response time required by service s ;
- $R_{\text{energy},s,k}^{\uparrow}$: amount that can be sustained for the required service duration, considering SOC.

19 Numerical Example: From FOR to FFR Capability

Suppose the current operating point is

$$P_0 = -0.5 \text{ MW}.$$

The FOR gives

$$R_{\text{network}}^{\uparrow} = 2.5 \text{ MW}.$$

The aggregated BESS power limit gives

$$R_{\text{power}}^{\uparrow} = 2.0 \text{ MW}.$$

Suppose only

$$1.8 \text{ MW}$$

can be reached within the required FFR response time.

Therefore,

$$R_{\text{dynamic,FFR}}^{\uparrow} = 1.8 \text{ MW}.$$

Suppose the current SOC is sufficient to sustain

$$2.0 \text{ MW}$$

for the required duration.

Then,

$$R_{\text{energy,FFR}}^{\uparrow} = 2.0 \text{ MW}.$$

The actual FFR capability is therefore

$$R_{\text{FFR}}^{\uparrow} = \min \{2.5, 2.0, 1.8, 2.0\}.$$

Hence,

$$R_{\text{FFR}}^{\uparrow} = 1.8 \text{ MW.}$$

Connection with the PhD framework

This illustrates an important extension of the FOR concept. The FOR can determine what is **network-feasible**. The ancillary-service requirements determine what part of that flexibility is also **fast enough and sustainable enough** for a specific service.

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20 From Available Flexibility to Service Activation and SOC Feedback

The frequency-service process can be understood as a sequence of three different concepts:

$$\text{Availability} \rightarrow \text{Activation} \rightarrow \text{Delivery}$$

followed by an update of the battery state.

1. Availability

Before any frequency event occurs, the model determines how much grid-feasible flexibility can be made available. For example,

$$R_{\text{FFR}}^{\uparrow} = 1.8 \text{ MW.}$$

This means

$$\text{“Up to 1.8 MW can be provided if the service is needed.”}$$

At this stage, **no frequency service has necessarily been activated**.

The BESS continues its normal local operation:

$$\begin{array}{ll} \text{PV surplus} & \rightarrow \text{BESS charges} \\ \text{Residual demand} & \rightarrow \text{BESS discharges} \end{array}$$

This self-consumption operation determines the actual BESS operating point and SOC.

2. Activation

If no qualifying frequency event occurs,

$$\text{No event} \rightarrow \text{no frequency-service activation.}$$

If frequency deviates sufficiently, part of the available flexibility is activated. For example,

$$R_{\text{FFR}}^{\uparrow} = 1.8 \text{ MW,}$$

but the actual required response may be only

$$\Delta P = 1.2 \text{ MW}.$$

Therefore,

$$\boxed{\text{Available flexibility} \neq \text{activated flexibility}.}$$

For a fast frequency service, this activation may be based directly on the measured frequency rather than waiting for a new OPF calculation. Thus,

$$\boxed{\text{OPF/FOR determines what can be available beforehand}}$$

while

$$\boxed{\text{fast control activates the required response when an event occurs}.}$$

3. Delivery

After activation, the physical DERs must actually provide the requested response at the PCC. For example, if

$$\Delta P = 1.2 \text{ MW},$$

the aggregated DERs must change the PCC active-power exchange by approximately

$$1.2 \text{ MW}$$

within the response time required by the service. Therefore,

$$\boxed{\text{Availability} \neq \text{Activation} \neq \text{Delivery}.}$$

4. Service Activation Changes the BESS SOC

If the service is activated, the BESS operates differently from the original self-consumption schedule. Consider a simple example. Suppose

$$E_b = 4 \text{ MWh},$$

the BESS discharges

$$P^{\text{dis}} = 1 \text{ MW},$$

for

$$15 \text{ min} = 0.25 \text{ h},$$

with

$$\eta_{\text{dis}} = 0.95.$$

The energy delivered to the grid is

$$E_{\text{grid}} = 1(0.25) = 0.25 \text{ MWh.}$$

The corresponding energy removed from the battery is approximately

$$E_{\text{battery}} = \frac{0.25}{0.95} = 0.263 \text{ MWh.}$$

Therefore,

$$\Delta SOC = \frac{0.263}{4} \approx 0.066,$$

or approximately

$$\boxed{\Delta SOC \approx 6.6\%}.$$

If the BESS starts at

$$SOC = 60\%,$$

the SOC after the service is approximately

$$SOC_{\text{after service}} = 53.4\%.$$

5. The Realized SOC Becomes the Next Initial Condition

The key consequence is that the SOC predicted by the original self-consumption schedule may no longer be correct. For example,

$$SOC_{k+1}^{\text{baseline}} = 58\%,$$

while the actual service activation results in

$$SOC_{k+1}^{\text{realized}} = 54\%.$$

Hence,

$$\boxed{SOC_{k+1}^{\text{realized}} \neq SOC_{k+1}^{\text{baseline}}}.$$

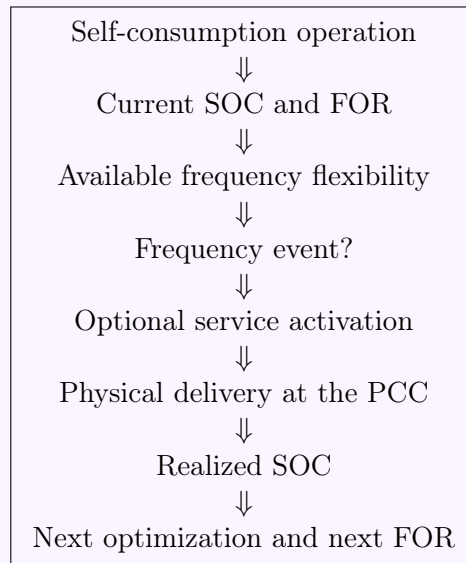
The next optimization must therefore start from the **realized SOC**:

$$\boxed{SOC_{k+1} = 54\%}.$$

This updated SOC then affects the flexibility available in the next interval and, consequently, the next FOR.

Connection with the PhD framework

The complete operational sequence is therefore



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21 Complete Connection with the PhD Framework

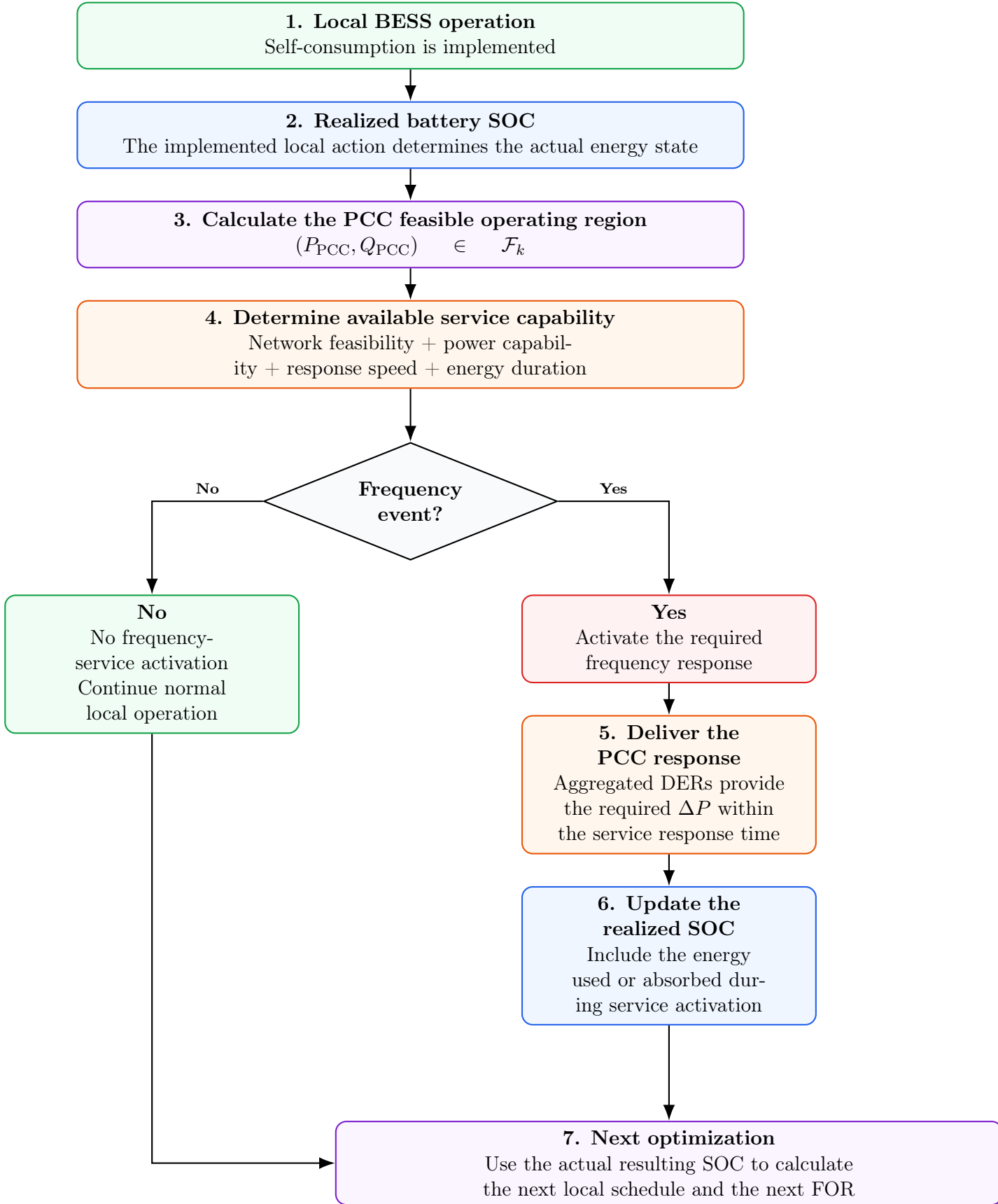


Figure 4: Operational sequence connecting self-consumption, realized SOC, the PCC feasible operating region, available frequency-service capability, optional service activation, and the next optimization.

22 A More Precise Interpretation of the Framework

The PhD framework can therefore be interpreted as having two main timescales.

22.1 Slower optimization layer

The optimization determines:

$$\boxed{\text{SOC} + \text{network conditions} + \text{device limits}}$$

and calculates

$$\boxed{\text{grid-feasible flexibility at the PCC.}}$$

This can happen before any frequency event.

22.2 Fast physical-control layer

If frequency deviates,

$$f(t) \neq f_0,$$

the fast controller activates

$$\Delta P(t)$$

from the flexibility that was made available. Therefore,

$$\boxed{\text{FOR} / \text{OPF} \rightarrow \text{prepares the feasible capability}}$$

while

$$\boxed{\text{frequency controller} \rightarrow \text{activates that capability.}}$$

23 Service-Specific Flexibility

This suggests an important distinction. The conventional FOR is

$$\mathcal{F}_k.$$

But not every point in

$$\mathcal{F}_k$$

is necessarily reachable within every service timescale. For example, a point may be feasible in steady state but impossible to reach in less than one second. Conceptually, one could therefore distinguish the subset of the FOR that is suitable for a service s :

$$\boxed{\mathcal{F}_k^{(s)} \subseteq \mathcal{F}_k.}$$

For an FFR service,

$$\mathcal{F}_k^{(\text{FFR})}$$

would contain only responses that are:

1. network-feasible;
2. device-feasible;
3. reachable within the FFR response time;
4. sustainable for the required duration.

Connection with the PhD framework

This provides a useful conceptual extension for the research:

FOR $\rightarrow$ service-specific feasible flexibility.

The FOR determines what the distribution system can physically support from a network perspective. The frequency-service requirements determine which part of that flexibility is suitable for FFR, FCR, FRR, or another ancillary-service product.

24 Where Different DERs Could Enter the PhD Framework

Suppose an aggregator manages several types of DERs. For example:

- BESS;
- controllable loads;
- EVs;
- flexible renewable generation;
- small dispatchable generators.

The aggregator does not necessarily need to use every DER for every frequency service. Instead, resources can be selected according to the service requirements. For example,

FFR $\rightarrow$ primarily very fast resources

while

longer restoration service $\rightarrow$ resources capable of sustaining power longer.

The disaggregation problem can therefore include not only

“Who should provide the requested power?”

but also

“Which DERs are technically suitable for this service timescale?”

25 A Simple Aggregator Example

Suppose an aggregator has:

$$\text{BESS capacity} = 1.5 \text{ MW},$$

$$\text{flexible EV charging} = 0.5 \text{ MW},$$

and

$$\text{small controllable generator} = 1.0 \text{ MW}.$$

The total static active-power flexibility appears to be

$$1.5 + 0.5 + 1.0 = 3.0 \text{ MW}.$$

But suppose only:

- the BESS can provide its full response in less than one second;
- EV charging can change by 0.3 MW quickly enough;
- the controllable generator is too slow for the FFR requirement.

Then the FFR capability is not

$$3.0 \text{ MW}.$$

Instead,

$$R_{\text{FFR}} = 1.5 + 0.3 = 1.8 \text{ MW}.$$

The generator may still be very useful later for frequency restoration. This demonstrates why

$$\text{total flexibility} \neq \text{FFR flexibility}.$$

26 Important Abbreviations

Term	Full name	Plain-language meaning
AS	Ancillary Service	A service used to support reliable grid operation in addition to normal energy delivery.
DER	Distributed Energy Resource	A resource connected mainly to the distribution grid, such as BESS, PV, EV, flexible load, or small generator.
BESS	Battery Energy Storage System	A battery that can charge, discharge, and provide grid flexibility.
RES	Renewable Energy Source	Renewable generation such as solar or wind.
TSO	Transmission System Operator	The operator responsible for the high-voltage transmission system.
DSO	Distribution System Operator	The operator responsible for secure operation of the distribution grid.
PCC	Point of Common Coupling	The interface at which the distribution grid exchanges active and reactive power with the upstream system.
SOC	State of Charge	The amount of energy currently stored in a battery.
FOR	Feasible Operating Region	The set of feasible active- and reactive-power exchanges at the PCC.
FFR	Fast Frequency Response	Very fast active-power response following a frequency disturbance.
PFR	Primary Frequency Response	The first controlled response used to help arrest a frequency deviation.
FCR	Frequency Containment Reserve	Reserve used to stabilize or contain frequency after an imbalance.
FRR	Frequency Restoration Reserve	Reserve used to restore frequency and scheduled system balance.
aFRR	Automatic Frequency Restoration Reserve	FRR activated automatically by a central frequency-restoration control system.
mFRR	Manual Frequency Restoration Reserve	FRR activated through manual or scheduled operator action rather than the automatic restoration controller.
RR	Replacement Reserve	Reserve used to replace previously activated restoration reserves.
AGC	Automatic Generation Control	Centralized control that changes participating resource setpoints to restore system balance.
RoCoF	Rate of Change of Frequency	How quickly frequency changes, normally expressed in Hz/s.
FCAS	Frequency Control Ancillary Services	Australian terminology for frequency-control service products.
NERC	North American Electric Reliability Corporation	North American reliability organization.
NESO	National Energy System Operator	Electricity system operator in Great Britain.
AEMO	Australian Energy Market Operator	Australian power-system and electricity-market operator.
ENTSO-E	European Network of Transmission System Operators for Electricity	Association of European transmission system operators.

Table 4: Main abbreviations used in the discussion.

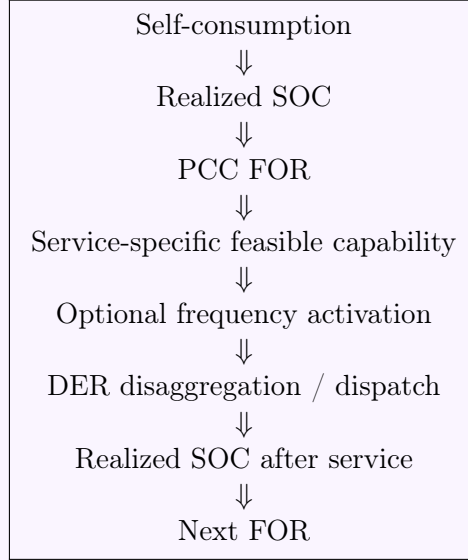
27 The Main Research Connection

Connection with the PhD framework

A useful research question emerging from this discussion is:

How much ancillary-service flexibility can an active distribution network safely make available at the PCC, considering the actual BESS energy state, distribution-network limits, and the response-time requirements of the requested service?

The complete operational chain becomes



28 Three Different Questions

The framework can now distinguish three questions.

28.1 Question 1: Is the operating point network-feasible?

This is addressed by the FOR:

$$(P, Q) \in \mathcal{F}_k.$$

28.2 Question 2: Can the DERs reach it quickly enough?

This depends on the ancillary-service response requirement:

$$t_{\text{response}} \leq t_s^{\max}.$$

28.3 Question 3: Can the DERs sustain it long enough?

This depends mainly on SOC and available energy:

$$E_{\text{available}} \geq E_{\text{service}}.$$

Therefore,

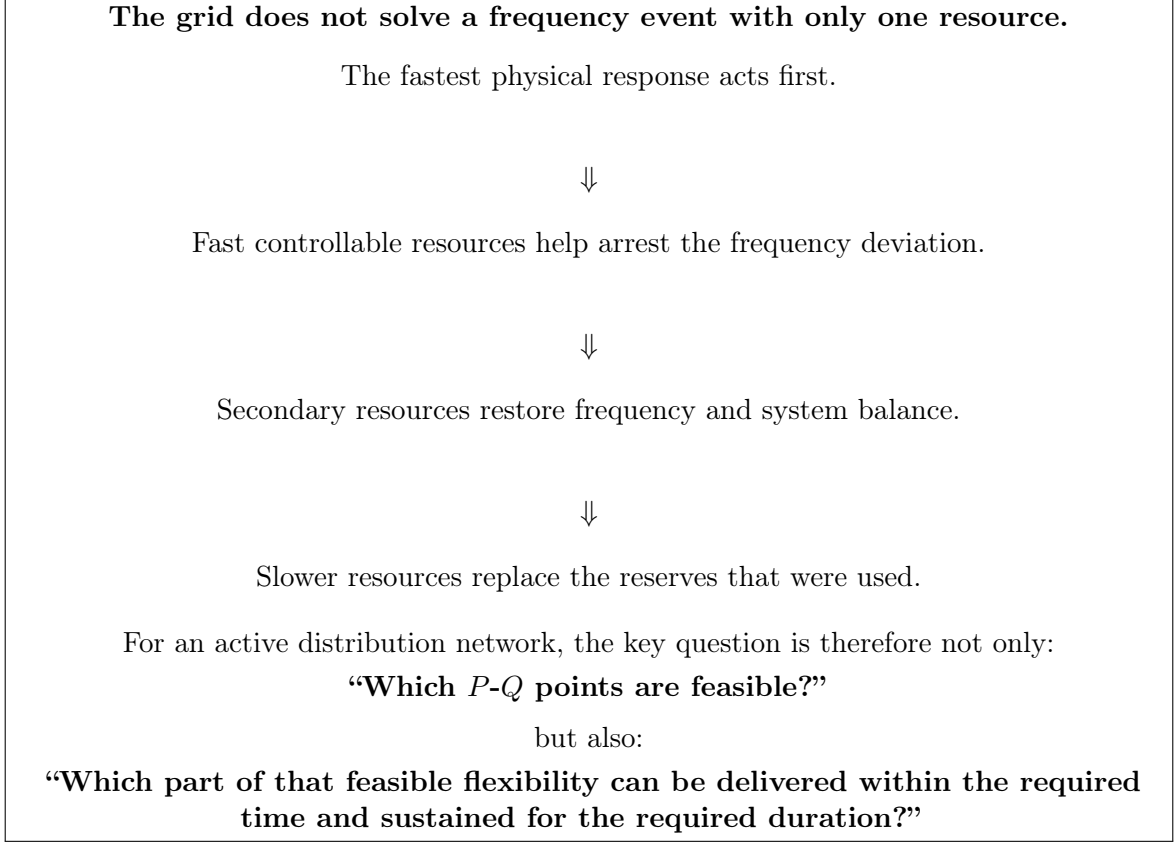
Network-feasible + fast enough + sustainable

are three different conditions.

29 Key Takeaways

1. Frequency falls when active-power generation becomes smaller than active-power demand.
2. Frequency rises when generation becomes larger than demand.
3. Different frequency-control actions operate at different timescales.
4. Inertia acts immediately and slows the initial frequency change.
5. FFR and primary response act rapidly to help arrest the deviation.
6. Secondary response helps restore frequency and system balance.
7. Replacement reserves release previously activated reserves and prepare the system for another event.
8. Different technologies are naturally suited to different response timescales.
9. There is no universal rule saying that only one specific generation technology can provide each service.
10. BESSs are especially attractive for fast frequency response because of their rapid active-power capability.
11. A resource must have sufficient power headroom, energy headroom, and network headroom.
12. The FOR provides information about network-feasible flexibility at the PCC.
13. Not every point inside a steady-state FOR is necessarily reachable in less than one second.
14. A service-specific capability should therefore also consider response speed and service duration.
15. Local self-consumption remains the normal BESS operation.
16. Frequency-service activation occurs only if the corresponding event or activation condition occurs.
17. If a service is activated, the resulting BESS dispatch changes SOC.
18. The realized SOC becomes the starting state for the next optimization.
19. The next FOR must therefore reflect the actual previous operation, including any ancillary-service activation.

Final Mental Picture



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